Matthew Maciejewski

Curriculum Vitae

✉ matt@mmaciejewski.com

Research Specialization

speech & audio processing, signal processing, deep learning, statistical modeling

Education

2014–2021 **Ph.D., Electrical Engineering**, *Johns Hopkins University*, Baltimore, MD.
Advisors: Sanjeev Khudanpur and Shinji Watanabe
Dissertation Topic: speech separation in noisy and reverberant conditions

2014–2017 **M.S., Electrical and Computer Engineering**, *Johns Hopkins University*, Baltimore, MD.

2010–2014 **B.S., Electrical and Computer Engineering**, *Carnegie Mellon University*, Pittsburgh, PA.

Experience

2023–Present **Research Scientist**, *Human Language Technologies Center of Excellence, Johns Hopkins University*, Baltimore, MD.
Research in speech technologies focused on audio captured by a distant microphone, including speaker diarization and speech enhancement.

2021–2023 **Research Scientist**, *Amazon Alexa AI*, Cambridge, MA.
Research in acoustic event detection and acoustic scene classification. Leveraged knowledge distillation and data mining to deploy improved event detectors with a lower on-device resource footprint. Mentored interns that led to return offers.

2014–2021 **Research Assistant**, *Department of Electrical and Computer Engineering, Johns Hopkins University*, Baltimore, MD.
Research in speech processing with emphasis on conversational speech, primarily far-field speech separation. This included speaker diarization, speaker recognition, and speech enhancement, and contributions to the Kaldi ASR toolkit.

2019–2019 **Research Intern**, *Mitsubishi Electric Research Laboratories (MERL)*, Cambridge, MA.
Worked on noisy and reverberant speech separation, releasing the WHAMR! corpus and improved performance through cascaded model architecture.

2016–2016 **Software Engineer Intern**, *Spoken Communications*, Seattle, WA.
Made improvements to the speech activity detection system and ported the speaker verification system to a new C++ framework.

2013–2014 **Undergraduate Researcher**, *Carnegie Mellon University*, Pittsburgh, PA.
Worked on dereverberation of speech signals in binaural recordings.

Professional Activities

Organizer	CHiME-8, CHiME-7
Session Chair	Interspeech 2025, Odyssey 2024
Reviewer	ICASSP, Interspeech, SLT, ASRU, DCASE, WASPAA, IWAENC, CHiME, Applied Acoustics, Computer Speech and Language, Speech Communication

Teaching Experience

Co-Instructor	520.666 Information Extraction *JHU* 2026
Guest Lecturer	601.667 Introduction to Human Language Technology *JHU* 2023–Present
	JSALT 2024 Summer School

Student Development

PhD Co-Advisor	Xavier Velez. *Audio Representation Learning*
Qualifying Project	Xiluo He. *Stream-Conditioned Multi-Talker ASR*. Published at ICASSP 2026
Masters Thesis	Dominik Klement. *Unsupervised Speech Enhancement*. Masters Thesis Award
Research Project	David Palzer
PhD Committee	Yiwen Shao
GBO Committee	Neha Verma, Zili Huang

Grants and Funding

Submitted	NSF Future CoRe *Native Phase Processing via Complex-Valued Neural Networks for Speech Enhancement and Acoustic Beamforming*

Awards

CHiME-8	Jury Prize for Task 2: NOTSOFAR

Publications

2027 S. Cornell, **M. Maciejewski**. *SNOAR: Self-Training with Noise Filtering, Observation Adding, and Remixing for Speech Denoising without Clean References*. (under review)

X. Velez, **M. Maciejewski**, S. Khudanpur. *Towards Integrating Dereverberation Into A Neural Audio Codec-Based Processing Chain*. (under review)

R. Kriz, D. Etter, A. Martin, C. Carpenter, D. Chakraborty, H. Recknor, R. Iranmanesh, **M. Maciejewski**, K. Murray, E. Yang, B. Van Durme, A. S. White, A. Yates, W. Walden. *MultiVENT-Raw: A Benchmark for Retrieval and Reasoning over Raw Videos*. (under review)

2026 **M. Maciejewski**, S. Cornell. *Ring Mixing with Auxiliary Signal-to-Consistency-Error Ratio Loss for Unsupervised Denoising in Speech Separation*. IEEE SLT. 2026

D. Klement, **M. Maciejewski**, S. Khudanpur, J. Černocký, L. Burget. *Analysing Adversarial Priors for Data-driven Unsupervised Speech Enhancement*. ISCA Interspeech. 2026

M. Maciejewski, S. Cornell. *Exploiting Noise Inseparability for Weakly-Supervised Discriminative Speech Denoising Using Noisy Targets*. IEEE IWAENC. 2026

M. Maciejewski. *Single-Microphone Audio Point Source Discriminative Localization from Reverberation Late Tail Estimation*. IEEE ICASSP. 2026

X. He, A. Polok, J. Villalba, T. Thebaud, **M. Maciejewski**. *Scaling Multi-Talker ASR with Speaker-Agnostic Activity Streams*. IEEE ICASSP. 2026

Z. Huang, **M. Maciejewski**, L.P. Garcia Perera, S. Watanabe, S. Khudanpur. *CSPB: Conversational Speech Processing Benchmark for Self-supervised Speech Models*. EACL. 2026

2025 **M. Maciejewski**. *Speaker Conditioning of Voice Activity Detection via Implicit Separation*. ISCA Interspeech. 2025

D. Palzer, **M. Maciejewski**, E. Fosler-Lussier. *End-to-End Diarization utilizing Attractor Deep Clustering*. ISCA Interspeech. 2025

2024 A. Polok, D. Klement, J. Han, Š. Sedláček, B. Yusuf, **M. Maciejewski**, M. Wiesner, L. Burget. *BUT/JHU System Description for CHiME-8 NOTSOFAR-1 Challenge*. Proc. CHiME-8. 2024

S. Cornell, T.J. Park, H. Huang, C. Boeddeker, X. Chang, **M. Maciejewski**, M. Wiesner, P. Garcia, S. Watanabe. *The CHiME-8 DASR Challenge for Generalizable and Array Agnostic Distant Automatic Speech Recognition and Diarization*. Proc. CHiME-8. 2024

M. Maciejewski, D. Klement, R. Huang, M. Wiesner, S. Khudanpur. *Evaluating the Santa Barbara Corpus: Challenges of the Breadth of Conversational Spoken Language*. ISCA Interspeech. 2024

D. Raj, M. Wiesner, **M. Maciejewski**, L.P. Garcia-Perera, D. Povey, S. Khudanpur. *On Speaker Attribution with SURT*. ISCA Odyssey. 2024

D. Palzer, **M. Maciejewski**, E. Fosler-Lussier. *Improving Neural Diarization Through Speaker Attribute Attractors and Local Dependency Modeling*. IEEE ICASSP. 2024

2023 S. Cornell, M. Wiesner, S. Watanabe, D. Raj, X. Chang, P. Garcia, **M. Maciejewski**, Y. Masuyama, Z.-Q. Wang, S. Squartini, S. Khudanpur. *The CHiME-7 DASR Challenge: Distant Meeting Transcription with Multiple Devices in Diverse Scenarios*. Proc. CHiME-7. 2023

M. Maciejewski, J. Shi, S. Watanabe, S. Khudanpur. *A Dilemma of Ground Truth in Noisy Speech Separation and an Approach to Lessen the Impact of Imperfect Training Data*. Computer Speech & Language. 2023

2021 **M. Maciejewski**. *Towards Single-Channel Speech Separation in Noise and Reverberation*. [Doctoral dissertation]. Johns Hopkins University. 2021

M. Maciejewski, S. Watanabe, S. Khudanpur. *Speaker Verification-Based Evaluation of Single-Channel Speech Separation*. ISCA Interspeech. 2021

M. Maciejewski, J. Shi, S. Watanabe, S. Khudanpur. *Training Noisy Single-Channel Speech Separation with Noisy Oracle Sources: A Large Gap and a Small Step*. IEEE ICASSP. 2021

2020 **M. Maciejewski**, G. Wichern, E. McQuinn, J. Le Roux. *WHAMR!: Noisy and Reverberant Single-Channel Speech Separation*. IEEE ICASSP. 2020

A. Arora, D. Raj, A.S. Subramanian, K. Li, B. Ben-Yair, **M. Maciejewski**, P. Żelasko, P. Garcia, S. Watanabe, S. Khudanpur. *The JHU Multi-Microphone Multi-Speaker ASR System for the CHiME-6 Challenge*. Proc. CHiME-6. 2020

2019 **M. Maciejewski**, G. Sell, Y. Fujita, L.P. Garcia-Perera, S. Watanabe, and S. Khudanpur. *Analysis of Robustness of Deep Single-Channel Speech Separation Using Corpora Constructed From Multiple Domains*. IEEE WASPAA. 2019

2018 N. Kanda, R. Ikeshita, S. Horiguchi, Y. Fujita, K. Nagamatsu, X. Wang, V. Manohar, N.E.Y. Soplin, **M. Maciejewski**, S.J. Chen, A.S. Subramanian, R. Li, Z. Wang, J. Naradowsky, L.P. Garcia-Perera, G. Sell. *The Hitachi/JHU CHiME-5 system: Advances in speech recognition for everyday home environments using multiple microphone arrays*. Proc. CHiME-5. 2018

G. Sell, D. Snyder, A. McCree, D. Garcia-Romero, J. Villalba, **M. Maciejewski**, V. Manohar, N. Dehak, D. Povey, S. Watanabe, and S. Khudanpur. *Diarization is Hard: Some Experiences and Lessons Learned for the JHU Team in the Inaugural DIHARD Challenge*. ISCA Interspeech. 2018

M. Maciejewski, D. Snyder, V. Manohar, N. Dehak, and S. Khudanpur. *Characterizing Performance of Speaker Diarization Systems on Far-field Speech using Standard Methods*. IEEE ICASSP. 2018

N. Ryant, E. Bergelson, K. Church, A. Cristia, J. Du, S. Ganapathy, S. Khudanpur, D. Kowalski, M. Krishnamoorthy, R. Kulshreshta, M. Liberman, Y.D. Lu, **M. Maciejewski**, F. Metze, J. Profant, L. Sun, Y. Tsaok, Z. Yum. *Enhancement and Analysis of Conversational Speech: JSALT 2017*. IEEE ICASSP. 2018

2016 E. Chodroff, **M. Maciejewski**, J. Trmal, S. Khudanpur, J. Godfrey. *New release of Mixer-6: Improved validity for phonetic study of speaker variation and identification*. ELRA LREC. 2016

2015 H. Hermansky, L. Burget, J. Cohen, E. Dupoux, N. Feldman, J. Godfrey, S. Khudanpur, **M. Maciejewski**, S.H. Mallidi, A. Menon, T. Ogawa, V. Peddinti, R. Rose, R. Stern, M. Wiesner, K. Veselý. *Towards machines that know when they do not know: Summary of work done at 2014 Frederick Jelinek Memorial Workshop*. IEEE ICASSP. 2015

H.M. Park, **M. Maciejewski**, C. Kim, R. Stern. *Robust automatic speech recognition in reverberation: onset enhancement versus binaural source separation*. The Journal of the Acoustical Society of America. 2015

2014 H.M. Park, **M. Maciejewski**, C. Kim, R. Stern. *Robust speech recognition in reverberant environments using subband-based steady-state monaural and binaural suppression*. ISCA Interspeech. 2014